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The herbicide glyphosate causes behavioral changes and alterations in dopaminergic markers in male Sprague-Dawley rat

Isela Hernández-Plata [a], Magda Giordano [a], Mauricio Díaz-Muñoz [b], Verónica M. Rodríguez [a,*]

[a] Departamento de Neurobiología Conductual y Cognitiva, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Boulevard Juriquilla 3001, Querétaro, Querétaro 76230, Mexico
[b] Departamento de Neurobiología Celular y Molecular, Instituto de Neurobiología, Universidad Nacional Autónoma de México, Boulevard Juriquilla 3001, Querétaro, Querétaro 76230, Mexico

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ABSTRACT

Glyphosate (Glyph) is the active ingredient of several herbicide formulations. Reports of Glyph exposure in humans and animal models suggest that it may be neurotoxic. To evaluate the effects of Glyph on the nervous system, male Sprague-Dawley rats were given six intraperitoneal injections of 50, 100, or 150 mg Glyph/kg BW over 2 weeks (three injections/week). We assessed dopaminergic markers and their association with locomotor activity. Repeated exposure to Glyph caused hypoactivity immediately after each injection, and it was also apparent 2 days after the last injection in rats exposed to the highest dose. Glyph did not decrease monoamines, tyrosine hydroxylase (TH), or mesencephalic TH+ cells when measured 2 or 16 days after the last Glyph injection. In contrast, Glyph decreased specific binding to D1 dopamine (DA) receptors in the nucleus accumbens (NAcc) when measured 2 days after the last Glyph injection. Microdialysis experiments showed that a systemic injection of 150 mg Glyph/kg BW decreased basal extracellular DA levels and high-potassium-induced DA release in striatum. Glyph did not affect the extracellular concentrations of 3,4-dihydroxyphenylacetic acid or homovanillic acid. These results indicate that repeated Glyph exposure results in hypoactivity accompanied by decreases in specific binding to D1-DA receptors in the NAcc, and that acute exposure to Glyph has evident effects on striatal DA levels. Additional experiments are necessary in order to unveil the specific targets of Glyph on dopaminergic system, and whether Glyph could be affecting other neurotransmitter systems involved in motor control.

1. Introduction

Glyphosate (Glyph) is a phosphonomethyl amino acid derivative used as active ingredient in some herbicides. In recent years, the commercialization of glyphosate herbicides has increased due to development and sowing of glyphosate-resistant seeds, such as corn, soybeans, canola, and cotton (Dill et al., 2008). In plants, glyphosate inhibits the synthesis of aromatic amino acids, a metabolic pathway absent in mammals. Thus, it was considered that glyphosate herbicides were not a risk for the health of mammals including humans (Williams et al., 2000). However, in recent years, reports of human exposure and animal models suggest that both the commercial mixture containing glyphosate and the active ingredient glyphosate could have neurotoxic effects. Regarding human studies, glyphosate has been detected in brain and cerebrospinal fluid after exposure to commercial mixtures, revealing that the active ingredient can cross the blood brain barrier (Menkes et al., 1991; Sato et al., 2011). In addition, structural MRI studies in a subject exposed to the commercial mixture of glyphosate showed modifications in the T2 signal in substantia nigra, periaqueductal gray and globus pallidus, revealing possible bilateral lesions (Barbosa et al., 2001). Abnormal EEG activity and a Parkinsonian syndrome characterized by limb tremor at rest, global akinesia and rigidity have been observed after occupational exposure and accidental ingestion of the commercial mixture of glyphosate (Barbosa et al., 2001; Malhotra et al., 2010; Wang et al., 2011).

* Corresponding author. Tel.: +52 442 238 1061; fax: +52 442 238 1046.
E-mail addresses: vermire@yahoo.com, vmrodriguez@comunidad.unam.mx (V.M. Rodríguez).

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Studies in the rat have shown that oral exposure to the commercial mixture of glyphosate (1%, w/v) during pregnancy was associated with increased enzymatic activity of isocitrate dehydrogenase in the brain of pregnant dams, while the activities of isocitrate dehydrogenase, glucose 6 phosphate dehydrogenase and malate dehydrogenase were found to be increased in fetal brain (gestational day 21) (Daruich et al., 2001). In adult male rats, chronic exposure to 10 mg/kg BW glyphosate (three intraperitoneal [i.p.] injections per week for 5 weeks) was associated with decreased mitochondrial content of cardiolipin in the substantia nigra and cerebral cortex, while the exposure to a combination of glyphosate, dimethoate and zineb was associated with increased mitochondrial levels of thiobarbituric acid reactive substances (TBARS) in the same brain areas (Astiz et al., 2009). Recently, it was found that glyphosate causes cell death in the dopaminergic cell line PC12 through apoptotic and autophagic mechanisms (Gui et al., 2012). Taken together, these studies suggest that glyphosate exposure may affect the dopaminergic system of humans and rats, probably by oxidative stress. However, there have been no systematic studies evaluating the effects of the repeated and acute exposure to glyphosate on the dopaminergic systems in an *in vivo* model.

The dopaminergic nigrostriatal and mesolimbic systems are involved in the control of movement and motivated behaviors, and they have been shown to be vulnerable to herbicides such as paraquat and atrazine (ATR) (Thiruchelvam et al., 2000; Rodríguez et al., 2013). For instance, male mice treated with 10 mg/kg paraquat plus 30 mg/kg maneb (i.p. injection, twice per week for 3 weeks) showed reduced motor activity immediately after treatment. On the 5th day after treatment, the tyrosine hydroxylase (TH) levels were reduced in the striatum. Also, TH-positive cells decreased in the substantia nigra pars compacta (SNpc) (Thiruchelvam et al., 2000). Recently, Rodríguez et al. (2013) found that adult male Sprague-Dawley rats exposed to 100 mg ATR/kg BW (three i.p. injections per week for 2 weeks) developed hypoactivity that lasted up to 5 days after each injection. They also found reduced striatal DA, DOPAC and homovanillic acid (HVA) on the 6th day after treatment. In addition, ATR exposure reduced the expression of mRNA for cytosolic thioredoxin in NAcc, and for TH and the dopamine transporter in midbrain, and increased the expression of mRNA for the vesicular monoamine transporter in midbrain. These reports emphasize the vulnerability of the dopaminergic nigrostriatal and mesolimbic systems to herbicide exposure.

Therefore, in the present study, we evaluated the integrity of the nigrostriatal and mesolimbic dopaminergic systems by assessing biochemical (monoamine content, TH levels and specific binding to D1 and D2-DA receptors) and histological (mesencephalic TH+ cells) dopaminergic markers and their relationship with spontaneous locomotor activity in male rats that were repeatedly or acutely exposed to Glyph.

2. Materials and methods

2.1. Animals

Male Sprague-Dawley rats of 250–300 g were obtained from Harlan Laboratories Inc. (Mexico City, Mexico). They were housed in a room under constant temperature (23 °C), 12-h/12-h inverted light-dark cycle (lights on at 21:00 h) and water and food *ad libitum*. Animals were habituated to the vivarium for at least one week before beginning the experiments. Subjects were cared for and treated according to the Norma Oficial Mexicana (SAGARPA NOM-062-ZOO-1999), which complies with the guidelines in the Institutional Animal Care and Use Committee Guidebook (NIH Publication 80-23, Bethesda, MD, USA, 1996), and the protocol was approved by the local Committee on Bioethics.

2.2. Chemicals

Glyphosate and reagents for high performance liquid chromatography were acquired from Sigma–Aldrich (St. Louis, MO, USA), and the reagents for western blot were obtained from BioRad (Hercules, CA, USA). [^{3}H] SCH23390 and [^{3}H] Spiperone were purchased from Perkin-Elmer (Boston, MA, USA).

2.3. Experimental design

We chose three doses 50, 100 or 150 mg Glyph/BW based on the lethal dose 50 (LD_{50}) for rats (235 mg Glyph/kg BW, i.p.). Each animal received six injections over 2 weeks (three injections/week) of saline solution (vehicle) or Glyph (Experiment 1, Fig. 1). Locomotor activity was recorded during 15 min before, and for the 3 h immediately after saline or Glyph administration. On the 2nd and 16th day post-treatment, locomotor activity was recorded for 15 min before injection of normal saline solution (0.9% NaCl), and for 3 or 24-h after this injection. After the locomotor activity recording, rats were sacrificed by decapitation, and blood samples collected from trunk and brain regions (striatum – STR and nucleus accumbens – NAcc) were dissected and frozen at −70 °C for later evaluation of monoamine levels (n = 10), TH levels (n = 10), and specific binding to the dopamine receptors 1 and 2 (D1 and D2) (n = 6); the remaining tissue was fixed in 4% paraformaldehyde in order to quantify the number of mesencephalic tyrosine hydroxylase (TH) positive cells (n = 5–7). Serum was extracted from blood samples and stored at 4 °C in order to evaluate systemic damage through the enzymatic activity of aspartate aminotransferase, lactate dehydrogenase and gamma-glutamyltransferase. It is important to note that when it was necessary additional groups of rats were included, and treated as indicated in Fig. 1. The size of sample is indicated in each figure legend. Glyphosate doses, administration, and time post-treatment were established considering the requirements of the Office of Prevention, Pesticides and Toxic Substances, United States Environmental Protection Agency (OPPTS 870.6200 Neurotoxicity Screening Battery available at the website http://nepis.epa.gov.

2.4. Evaluation of the effects of Glyph on DA release in STR and NAcc

We analyzed the acute effects of glyphosate on DA release through simultaneous microdialysis in the STR and NAcc (Experiment 2, Fig. 1). Five to six rats per group were anesthetized with isoflurane (PISA, Guadalajara, Jalisco, Mexico) using an anesthesia system (SBH Scientific Designs Inc., Windsor, ON, Canada). The flowmeter was adjusted to 2 L/min for 1.5–2% (v/v) anesthetic gas. The pressure was maintained at 70 psi, and the body temperature was kept at 37 °C. The animals were placed on a stereotaxic frame (Kopf, Tujunga, CA, USA), and the cranial skin and muscle were removed to identify Bregma. The coordinates for STR were: AP: 1, ML: 2.5, DV: −6, and for NAcc: AP: 1.2, ML: 2.2, DV: −7.6 according to the atlas of Paxinos and Watson (2005). Probes were inserted in trephine holes for STR or NAcc in the right or left hemispheres, in an alternating pattern. Microdialysis probes (CMA/11, CMA/Microdialysis, Solna, Stockholm, Sweden) were 0.38 mm in diameter, 14 mm in length, and had an internal volume of 1 μL. The membrane had a diameter of 0.24 mm and a length of 2 mm for STR or 1 mm for NAcc. After inserting the probes, a Ringer solution (147 mM NaCl, 4 mM KCl, 1.2 mM $CaCl_2$, 1 mM $MgCl_2$) was perfused at a flow of 2.5 μL/min. No dialysate was collected during the first hour. Then, we collected 18 samples at 20-min intervals in vials containing 4 μL 0.1 N perchloric acid. The first

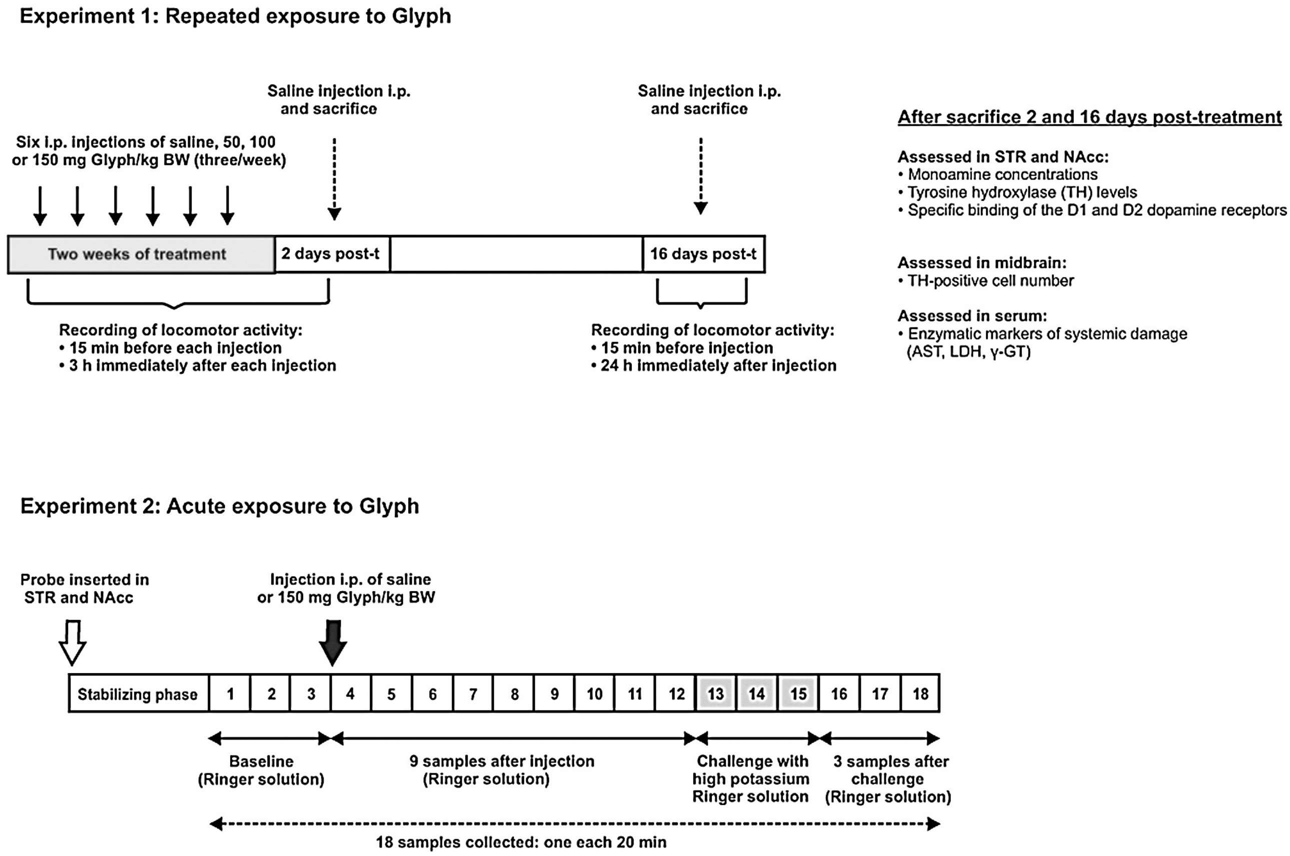

Fig. 1. Experimental design of the repeated and acute exposures to glyphosate (Glyph) in male rats.

three samples were considered as baseline. At the start of the fourth sample, rats received an i.p. injection of saline solution ($n = 6$) or 150 mg Glyph/kg BW ($n = 6$). Nine additional samples were then collected. From the 13th to 15th samples, a high-potassium Ringer solution (91 mM NaCl, 60 mM KCl, 1.2 mM $CaCl_2$ and 1 mM $MgCl_2$) was perfused through the probe. The last three samples (from 16th to 18th) were collected with Ringer solution. At the end of the microdialysis, the rats were decapitated, and the brain was removed and stored in 4% paraformaldehyde. Once fixed, the brain was frozen and sectioned in a freezing microtome (Leica SM 2000R, Wetzlar, Hesse, Germany) to verify the probe location in the STR and NAcc. Microdialysis samples were analyzed by high performance chromatography using an electrochemical detector (HPLC-DE, Perkin-Elmer, Waltham, MA, USA).

2.5. Locomotor activity

In order to assess the effects of glyphosate exposure on locomotor activity we used the protocol of Rodríguez et al. (2013). Briefly, this activity was evaluated using an automatic recording system (Digiscan Animal Activity Monitors – Accuscan Inc., Columbus, OH, USA). Animals were placed in acrylic chambers (40 cm × 40 cm × 40 cm) surrounded by horizontal and vertical infrared beams. During the course of the experiment, locomotor activity was recorded for 15 min before (exploratory activity) and for the 3-h period immediately after (spontaneous locomotor activity) the administration of Glyph (50, 100 or 150 mg Glyph/kg BW) or saline solution (vehicle). On the 2nd and 16th day post-Glyph treatment, all animals received a saline solution injection, and locomotor activity was evaluated for 15 min before and for 3-h on the 2nd day post-Glyph or for 24-h on the 16th day post-Glyph. Food and water were available throughout the 24-h of recording. In the present study, we selected three variables that represent the exploratory and ambulatory movements of the animals: total distance (distance traveled [in cm] by the animal in a given period), vertical activity or rearing (total number of beam interruptions that occurred in the vertical sensor during a given period) and stereotypy counts (number of beam breaks that occurred during repetitive movements such as grooming). The 3-h recording was divided into 15-min segments, and the 24-h recording was divided into 1-h segments.

2.6. Determination of monoamines and their metabolites

We assessed the effects of Glyph exposure on the total content of monoamines in the STR and NAcc following the protocol by Rodríguez et al. (2013). The results were reported for tissue as ng/mg tissue protein and ng/mL for microdialysates.

2.7. Tyrosine hydroxylase levels

TH levels were evaluated by Western blot; briefly, a portion of STR or NAcc was weighed and sonicated individually (1:10) in lysis buffer according to Pan et al. (2006) with slight modifications (50 mM Tris–HCl, 50 mM NaCl, 5 mM EDTA, 1 mM EGTA, 1% Triton X-100, 0.1% SDS, 10% glycerol and 1% of protease and phosphatase cocktail inhibitors [Sigma–Aldrich] at pH 7.5). Then, proteins were quantified using the Lowry DC assay (BioRad, Hercules, CA, USA). Each sample (5 μg of protein) was separated by electrophoresis on 12% SDS-PAGE and transferred to PVDF membranes using a semidry blotting apparatus (BioRad, Hercules, CA, USA). The membranes were blocked with 5% Blotto, non-fat dry milk (Santa

Cruz Biotechnology, Inc., Dallas, TX, USA) in TBS (100 mM Tris Base, 150 mM NaCl). Membranes were then incubated with anti-TH antibody (1:5000, Millipore, Temecula CA, USA) overnight at 4 °C, followed by anti-β-tubulin antibody (12G10, 1:2000, Developmental Studies Hybridoma Bank, University of Iowa, IA, USA). Then, we added the anti-rabbit and anti-mouse secondary antibodies conjugated to horseradish peroxidase (1:5000, Cell Signaling, Danvers, MA, USA). Finally, detection was performed with ECL-Plus kit (GE,

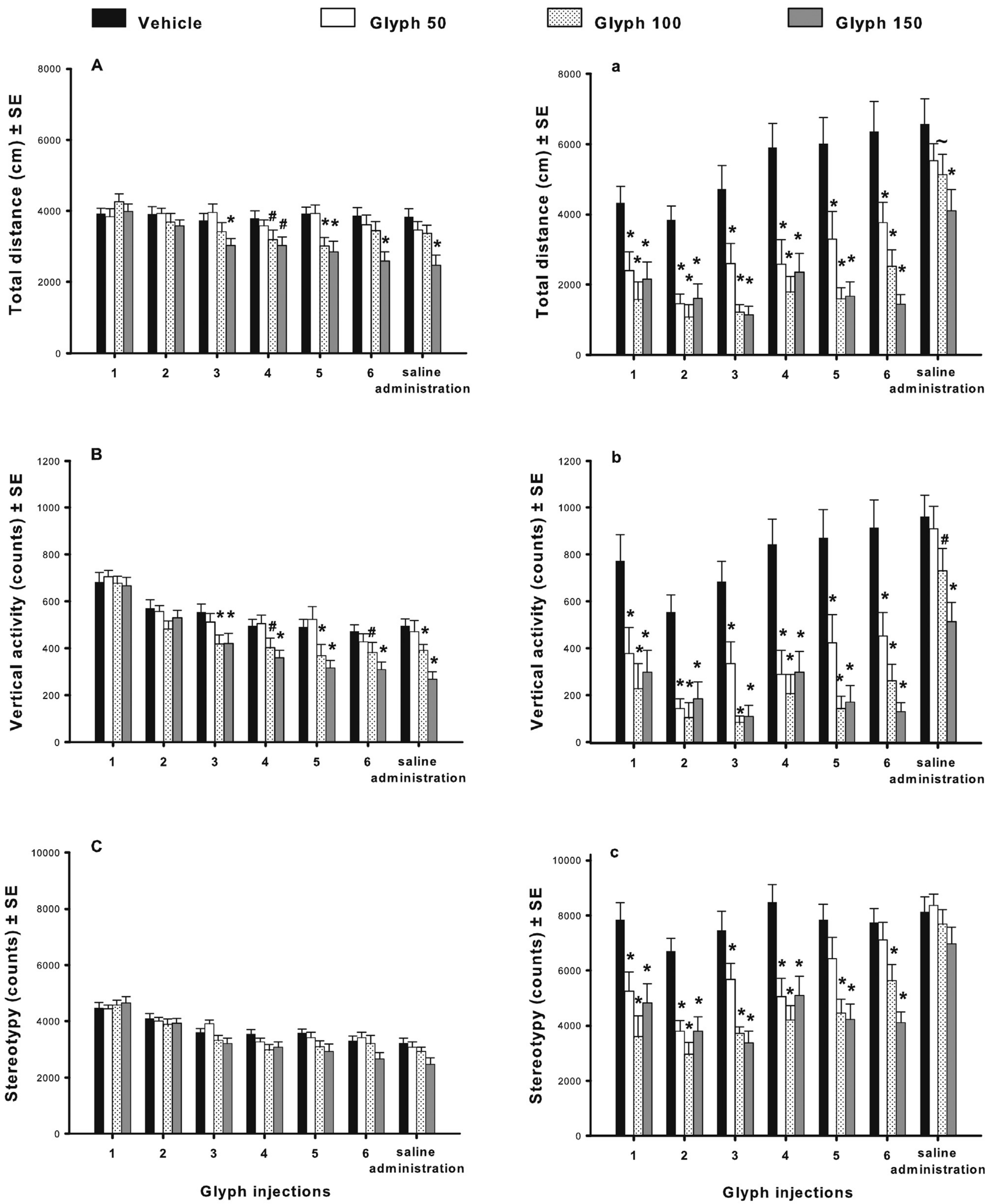

Fig. 2. Locomotor activity recorded for 15 min (A, B and C) before, and for the 3-h period immediately after (a, b and c) the i.p. injection of saline solution (vehicle), 50, 100 or 150 mg Glyph/kg BW. The exposure regime included three injections/week for 2 weeks and one saline injection 2 days after the last Glyph injection (n = 16–17). a, b and c represent the average sum of 3 h of recording. *Different from vehicle, $p < 0.05$; $^{\#}p$ = 0.05–0.07; $^{\sim}p < 0.1$.

Buckinghamshire, UK), and membranes were scanned using a STORM 860 (Amersham, Sunnyvale, CA, USA). Images were analyzed using Image J ver. 5.0 (NIH, Baltimore, MD, USA).

2.8. Binding assay for D1 and D2-DA receptors

Additional groups of rats were treated as shown before (Fig. 1, $n = 6$ per group). In these animals we recorded locomotor activity as described before, and collected tissue for the DA receptors radioligand assay. After treatment, STR and NAcc were removed from both hemispheres, weighed and stored at −70 °C until they were individually processed according to Levant (2007) with some modifications. STR or NAcc from each animal was homogenized in ice-cold 50 mM Tris–HCl buffer (1:20 w/v, pH 7.4) containing 1% protease and phosphatase cocktail inhibitors (Sigma–Aldrich). Thereafter, each homogenized sample was divided in half to assess either D1 or D2-DA receptors. All of the samples were centrifuged at 42,000 × g at 4 °C for 20 min. Then, the supernatant was

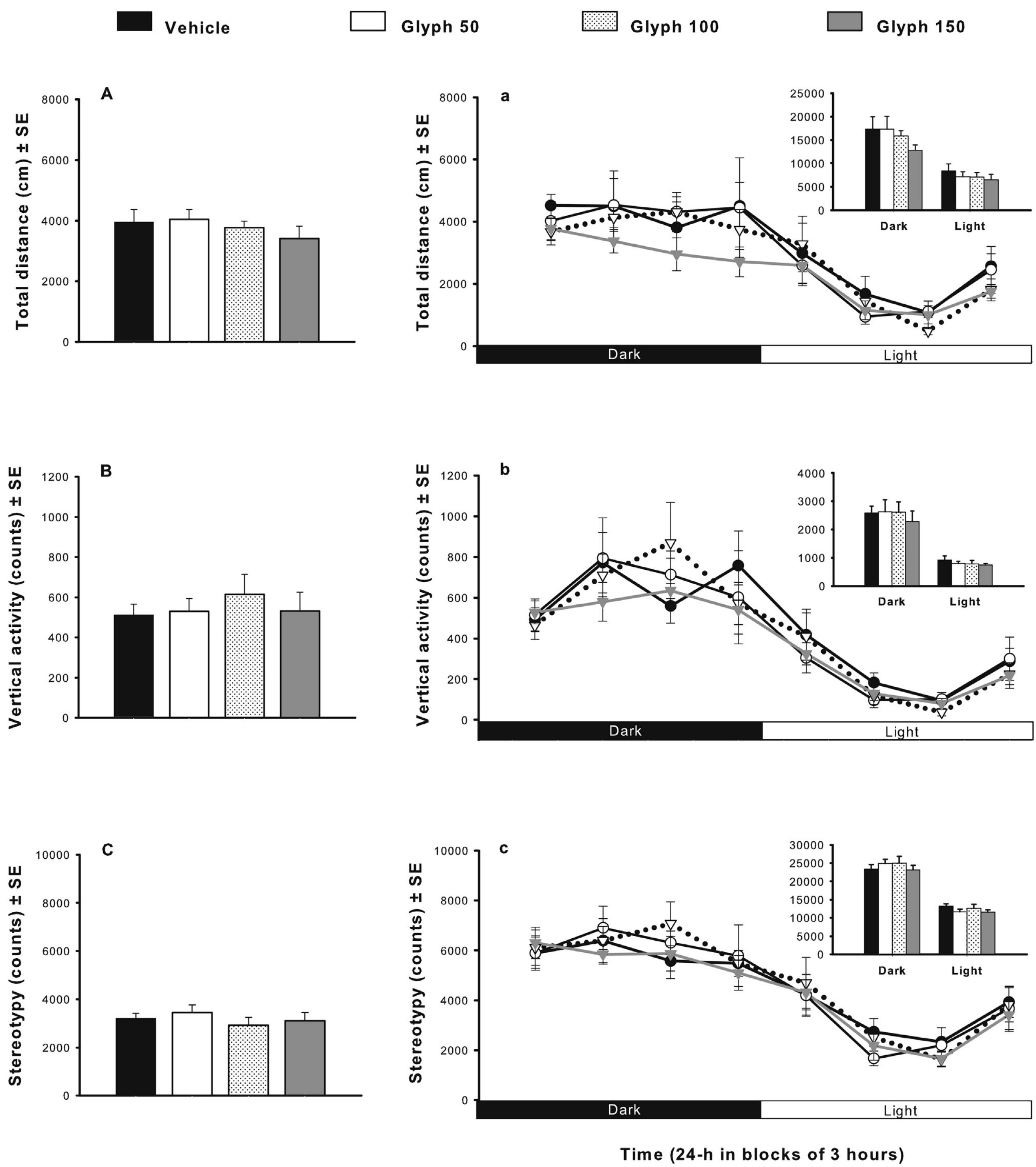

Fig. 3. Locomotor activity recorded 16 days after the last injection of vehicle, 50, 100 or 150 mg Glyph/kg BW. Exploratory activity was recorded 15 min before the i.p. saline injection (A, B and C) and for one 24-h light/dark cycle immediately after the i.p. saline injection (a, b and c). Insets show the average sum for dark and for light phase ($n = 6–7$).

Table 1
Body weight of animals treated with glyphosate.

Group	2 days post-treatment	16 days post-treatment
Vehicle	316 ± 7	377 ± 17
Glyph 50	314 ± 7	374 ± 14
Glyph 100	305 ± 6	365 ± 11
Glyph 150	310 ± 5	376 ± 10

Values are mean (g) ± SE. $n = 10$ and $n = 6$–7 for 2 and 16 days post-treatment, respectively.

discarded, and the pellet was resuspended using the same buffer and centrifugation conditions. Then, the supernatant was removed, and the pellet was resuspended in ice-cold assay buffer (3 mL/80 mg initial weight of tissue) for SCH23390 (50 mM Tris–HCl, 50 mM Tris base, 120 mM NaCl, 5 mM KCl, 2 mM $CaCl_2$, 1 mM $MgCl_2$, pH 7.4) or for spiperone (50 mM Tris–HCl, 50 mM Tris base, 5 mM KCl, 2 mM $CaCl_2$, 2 mM $MgCl_2$, pH 7.4) in order to assess D1 or D2-DA receptors, respectively. Finally, each sample was aliquoted and stored at −70 °C. Proteins were quantified by the Lowry DC assay (BioRad, Hercules, CA, USA).

To measure specific binding to D1-DA receptors in STR and NAcc, we used the specific ligand [^{3}H] SCH23390 (82.9 Ci/mmol, PerkinElmer, Kd 0.1–0.5 nM for the striatal and accumbal membrane assay). With each sample we ran two reactions, each in triplicate, to measure total binding and unspecific binding. The assay buffer to measure total binding of SCH23390 included 100 μL of 1.5 nM [^{3}H] SCH23390 for a final concentration of 0.3 nM, and 35 μg of tissue protein per sample. The unspecific binding assay buffer contained 100 μL of 1.5 nM of [^{3}H] SCH23390, 100 μL of 5 μM (+)-butaclamol for a final concentration of 1 μM and 35 μg tissue protein per sample. The final reaction volume was adjusted to 500 μL with assay buffer for SCH23390.

In order to assess binding to the D2-DA receptors in STR and NAcc, we used the specific ligand [^{3}H] spiperone (83.7 Ci/mmol, PerkinElmer, Kd 0.05–0.3 nM for the striatal and accumbal membrane assay). The procedure and preparation of the reactions were similar to that for D1-DA receptors except that we used 8.33 μL of 6 nM [^{3}H] spiperone for a final concentration of 0.1 nM.

All reactions were agitated and incubated for 90 min at room temperature. The reactions were stopped by rapid filtration through Whatman type C glass fiber filters (Whatman, Little Chalfont, Buckinghamshire, UK), and the filters were then washed five times with 3 mL of ice-cold wash buffer (50 mM Tris–HCl, 50 mM Tris base, pH 7.4). Filters were allowed to dry in scintillation vials for 1 h and then incubated overnight in 5 mL of scintillation fluid (13.55 mM 2,5-diphenyloxazole, 25.7% Triton™ X-100, 3.7% ethyleneglycol, 10.6% ethanol and 60% xylene). Finally, the samples were counted using a liquid scintillation counter (LS 6500, Beckman-Coulter, USA). The D1 and D2 binding was calculated as total binding minus unspecific binding. The results are expressed in nmol/mg protein.

Table 3
Content of monoamines in striatum and nucleus accumbens of glyphosate-treated rats measured 2 and 16 days post-treatment.

Group	DA	DOPAC	5-HT	5-HIAA	DOPAC/DA
2 days post-treatment					
Striatum					
Vehicle	100 ± 10	100 ± 13	100 ± 8	100 ± 8	100 ± 18
Glyph 50	109 ± 13	103 ± 19	105 ± 7	108 ± 9	93 ± 20
Glyph 100	121 ± 16	117 ± 17	94 ± 12	110 ± 12	102 ± 19
Glyph 150	118 ± 14	111 ± 23	109 ± 6	108 ± 8	95 ± 19
Nucleus accumbens					
Vehicle	100 ± 19	100 ± 20	100 ± 9	100 ± 14	100 ± 21
Glyph 50	132 ± 13	92 ± 14	111 ± 6	110 ± 12	58 ± 7
Glyph 100	86 ± 17	80 ± 17	112 ± 12	90 ± 17	92 ± 19
Glyph 150	119 ± 18	90 ± 16	121 ± 7	97 ± 13	74 ± 15
16 days post-treatment					
Striatum					
Vehicle	100 ± 5	100 ± 7	100 ± 10	100 ± 13	100 ± 9
Glyph 50	105 ± 11	116 ± 16	121 ± 16	125 ± 24	110 ± 12
Glyph 100	103 ± 15	110 ± 17	138 ± 28	133 ± 30	114 ± 17
Glyph 150	111 ± 12	115 ± 6	176 ± 42	136 ± 20	113 ± 15
Nucleus accumbens					
Vehicle	100 ± 19	100 ± 23	100 ± 8	100 ± 13	100 ± 17
Glyph 50	81 ± 10	79 ± 17	88 ± 9	86 ± 10	101 ± 17
Glyph 100	111 ± 27	102 ± 17	92 ± 15	90 ± 15	133 ± 22
Glyph 150	94 ± 9	105 ± 15	81 ± 7	85 ± 8	126 ± 20

Data expressed in percent of control ± SE. $n = 9$–10 in both experiments. DA, dopamine; DOPAC, dihydroxyphenylacetic acid; HVA, homovanillic acid; 5-HT, serotonin; 5-HIAA, 5-hydroxyindolacetic acid. Absolute values (ng/mg protein ± SE) for vehicle groups: 2 days post-treatment STR: DA = 84 ± 8, DOPAC = 6 ± 1, 5-HT = 6 ± 0.5, 5-HIAA = 4 ± 0.3, DOPAC/DA = 0.08 ± 0.01. NAcc: DA = 42 ± 8, DOPAC = 8 ± 2, 5-HT = 13 ± 1, 5-HIAA = 4 ± 0.9, DOPAC/DA = 0.2 ± 0.04. 16 days post-treatment STR: DA = 263 ± 14, DOPAC = 19 ± 1, 5-HT = 16 ± 2, 5-HIAA = 13 ± 2, DOPAC/DA = 0.07 ± 0.007. NAcc: DA = 123 ± 24, DOPAC = 8 ± 2, 5-HT = 26 ± 2, 5-HIAA = 14 ± 2, DOPAC/DA = 0.06 ± 0.01.

2.9. Tyrosine hydroxylase-positive cell number in substantia nigra and ventral tegmental area

To test if Glyph exposure modified the number of dopaminergic cells, we quantified the TH-positive cell number in the SN (substantia nigra pars compacta – SNpc, substantia nigra pars lateralis – SNpl, substantia nigra pars reticulata – SNpr) and VTA (ventral tegmental area – VTA, parabrachial pigmented nucleus – PBP, paranigral nucleus – PN, rostral linear nucleus raphe – RLi) of rats from each Glyph group at 2 and 16 days post-treatment. The immunocytochemical and cell counts were based on the methods described by Hoffman et al. (2008) and Bardullas et al. (2013). Midbrains were sectioned in the coronal plane into three series of 30-μm sections with a microtome (Leica SM 2000R, Wetzlar, Germany). Sections were observed using a 20× objective of a Leica DMLB optic microscope (Leica Microsystems, Wetzlar, Germany). For each animal, two observers blind to group assignment quantified TH-positive cells in all sections from Bregma −4.80 to −6.30 according to Paxinos and Watson (2005).

Table 2
Serum activities of lactate dehydrogenase, aspartate aminotransferase and gamma-glutamyltransferase, 2 and 16 days post-treatment.

Group	2 days post-treatment			16 days post-treatment		
	LDH	AST	γ-GT	LDH	AST	γ-GT
Vehicle	2137 ± 435	419 ± 77	131 ± 17	3589 ± 345	249 ± 29	136 ± 26
Glyph 50	3359 ± 1004	453 ± 68	110 ± 17	3407 ± 626	322 ± 37	208 ± 45
Glyph 100	4176 ± 733	454 ± 77	172 ± 37	4830 ± 446	371 ± 60	171 ± 27
Glyph 150	3238 ± 733	361 ± 36	101 ± 19	5093 ± 891	396 ± 41	158 ± 65

Values are mean (IU/L) ± SE. $n = 6$ in both experiments. LDH, lactate dehydrogenase; AST, aspartate aminotransferase and γ-GT, gamma-glutamyltransferase.

2.10. Evaluation of systemic biochemical markers of damage

In order to evaluate the possible systemic alterations induced by Glyph, we assessed, at 2 and 16 days post-treatment, the activity of three enzymatic markers of hepatic, kidney and heart damage in serum: aspartate aminotransferase (AST), lactate dehydrogenase (LDH) and gamma-glutamyltransferase (γ-GT) following the kit vendor's instructions (Wiener Lab Provides, Rosario, Argentina).

2.11. Statistical analysis

To evaluate the general effects of Glyph on locomotor activity, we used a repeated measure ANOVA (group $\times$ injection number), followed by one-way ANOVA (group) and LSD-Fisher *post hoc* tests.

For concentration of monoamines, TH levels, binding assay and enzymatic activity we used one-way ANOVAs and the LSD-Fisher *post hoc* test. For TH-positive cell number we used the Kruskal Wallis test. Finally, repeated measure ANOVAs and unpaired Student's *t*-tests were used to evaluate the microdialysis samples. The software StatView ver 5.0 was used in all analyses. Normality and homogeneity of variances were verified in all data. Statistical significance was defined as $p < 0.05$.

3. Results

3.1. Experiment 1. Effects of repeated exposure to Glyph 2 and 16 days post-treatment

3.1.1. Glyph exposure decreases exploratory behavior

Vehicle and Glyph groups displayed similar levels of exploratory activity before injections 1 and 2. However, exploratory activity was decreased in the animals treated either 100 or 150 mg Glyph/kg BW before rats received injection 3, and this hypoactivity lasted throughout most of the experiment, as shown in Fig. 2A and B. There were significant effects of treatment on total distance and vertical activity ($F_{(3,61)} = 5.49–7.46$, $p < 0.05$), injection number ($F_{(6,366)} = 7.522–45.77$, $p < 0.05$) and a significant interaction treatment $\times$ injection number ($F_{(18,366)} = 2.231–1.723$, $p < 0.05$), whereas only injection number had a significant effect ($F_{(6,366)} = 57.29$, $p < 0.0001$) on stereotypy counts.

3.1.2. Glyph exposure decreases spontaneous locomotor activity

Total distance, vertical activity and stereotypy counts decreased immediately after each Glyph administration (Fig. 2a–c) and this reduction in total distance traveled and vertical activity was still observed after a saline injection in the animals that received 150 mg Glyph/kg BW. Total distance, vertical activity and stereotypy counts showed effects of

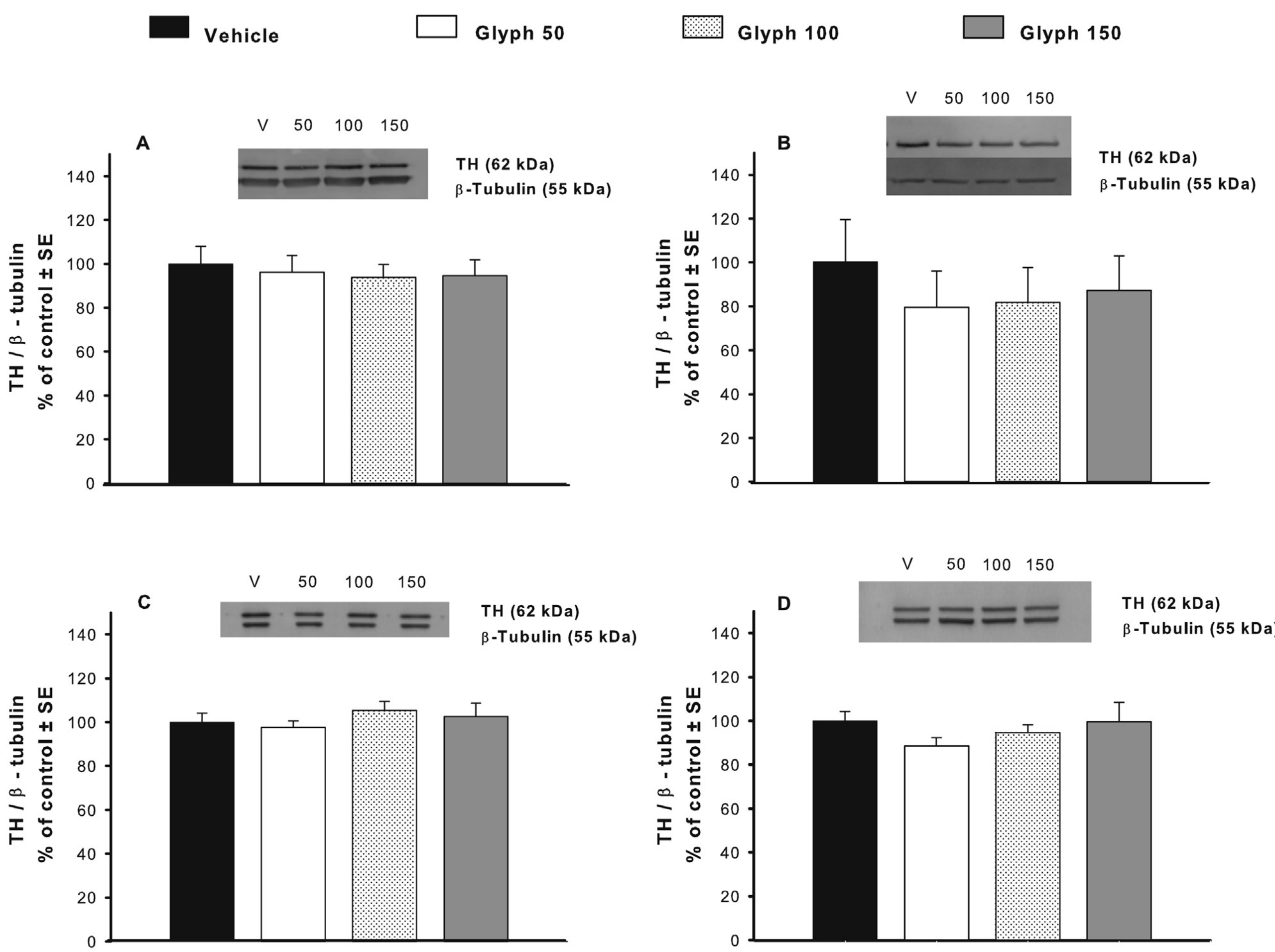

Fig. 4. Tyrosine hydroxylase (TH) levels in STR and NAcc of Glyph-treated rats 2 and 16 days after treatment ended expressed as percentage of the control group (vehicle). TH levels in STR (A) and NAcc (B) at 2 days post-treatment. TH levels in STR (C) and NAcc (D) at 16 days post-treatment $n = 10$. V, vehicle. Absolute values of TH/β-tubulin for vehicle group at 2 days post-treatment: STR = 0.90 $\pm$ 0.07, NAcc = 3.45 $\pm$ 0.7, and vehicle group at 16 days post-treatment: STR = 1.11 $\pm$ 0.05, NAcc = 0.85 $\pm$ 0.04.

treatment ($F_{(3,60)}$ = 22.23–36.14, $p < 0.0001$) and injection number ($F_{(6,360)}$ = 15.56–20.54, $p < 0.0001$). Only stereotypy counts showed interaction effects (treatment × injection number) ($F_{(18,360)}$ = 2.401, p = 0.0012).

In order to understand the acute effects of Glyph administration on locomotor activity, we exhaustively analyzed total distance traveled immediately after of the first injection of 150 mg Glyph/kg BW and saline solution (Supplementary Fig. 1B). There were treatment and time effects on total distance ($F_{(1,\ 31)}$ = 10.63, p = 0.002, $F_{(11,341)}$ = 13.6, $p < 0.0001$, respectively), which decreased by approximately 50% immediately after the administration of 150 mg Glyph/kg BW (sample 1: $t_{(31)}$ = 3.07, p = 0.004). Hypoactivity persisted, especially in samples 5, 6 and 9 ($t_{(31)}$ = 2.05, 2.44, 2.32 $p < 0.05$, respectively). Samples 11 and 12 showed only a trend to decrease ($t_{(31)}$ = 1.72, 1.87, $p < 0.1$, respectively).

3.1.3. Exploratory behavior and spontaneous locomotor activity did not show any change 16 days after Glyph exposure

Exploratory locomotor activity recorded 15 min before saline injection on the 16th day after Glyph treatment was similar in all experimental groups (Fig. 3A–C), just as was the spontaneous locomotor activity assessed during one 24-h dark/light cycle after saline injection (Fig. 3a–c).

3.1.4. Repeated Glyph exposure does not alter body weight or systemic biochemical markers of damage (LDH, AST and γ-GT)

Body weight, and the activity of LDH, AST and γ-GT were not affected by Glyph treatment (Tables 1 and 2).

3.1.5. Repeated Glyph exposure does not alter the content of monoamines or TH enzyme levels in STR and NAcc at 2 and 16 days post-treatment

Concentrations of DA, serotonin and their metabolites in STR or NAcc assessed 2 and 16 days post-treatment were similar in all groups (Table 3). The same result was observed with regard to TH levels in STR and NAcc (Fig. 4).

3.1.6. Repeated Glyph exposure decreases specific binding of the antagonist [^{3}H] SCH23390 to D1-DA receptors in NAcc 2 days post-treatment

The specific binding of [^{3}H] SCH23390 to D1-DA receptors in NAcc decreased in the animals exposed to 50, 100 and 150 mg Glyph/kg BW (decreases of 18%, 30% and 40%, respectively) measured 2 days post-treatment ($F_{(3,20)}$ = 8.71, p = 0.0007; Fig. 5C). In contrast, the specific binding of [^{3}H] Spiperone to D2-DA receptors was similar in all groups (Fig. 5D). In the STR, Glyph exposure caused no differences in specific binding of [^{3}H] SCH23390 or [^{3}H] Spiperone (Fig. 5A and B, respectively). Importantly, there were significant correlations between specific binding to D1-DA receptors in the NAcc and mean total distance traveled, and between binding to D1-DA receptors and mean vertical activity (r = 0.62, 0.68, $p < 0.01$, respectively, Fig. 7). The correlation between specific binding to D1-DA receptors in the NAcc and stereotypy counts was not significant (r = 0.37, p = 0.06) (Fig. 7). Specific binding of [^{3}H] SCH23390 and [^{3}H] Spiperone in the STR or NAcc (Fig. 6) was similar in all groups when measured on the 16th day after treatment.

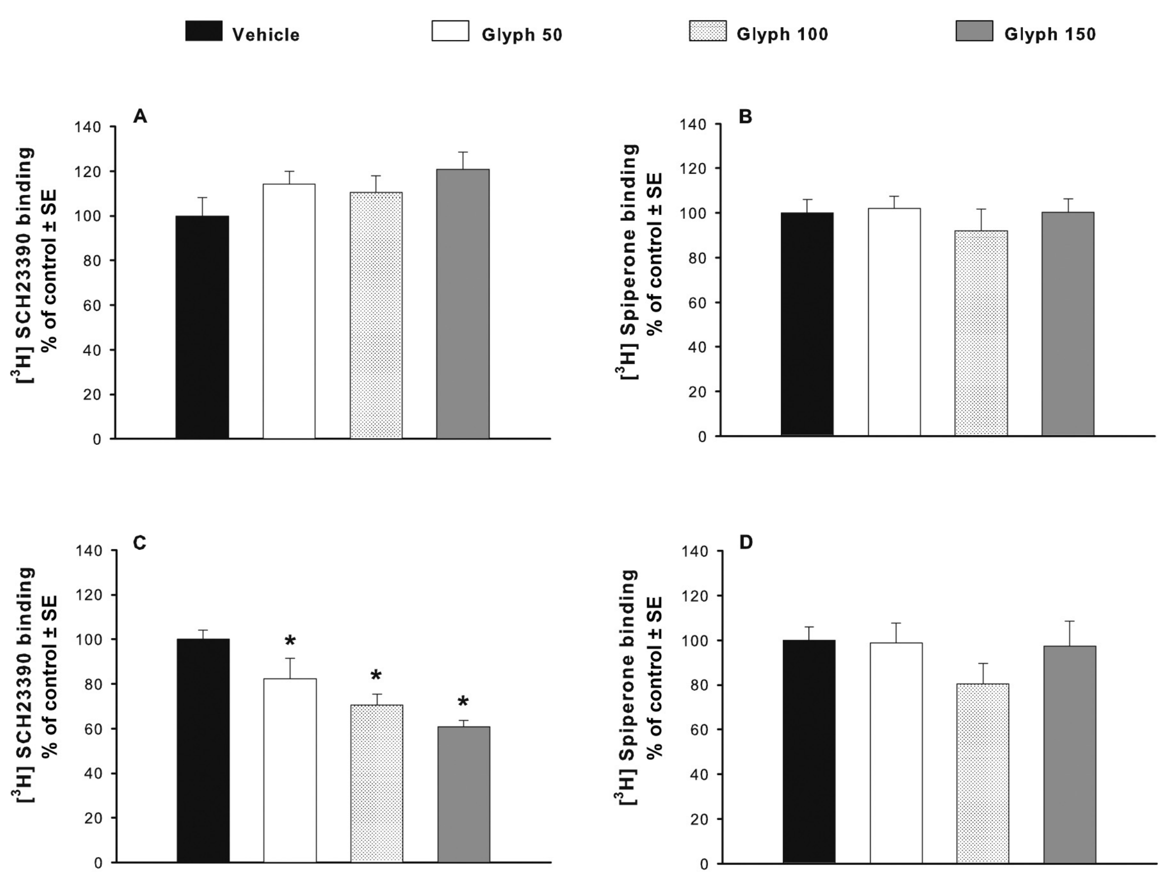

Fig. 5. Specific binding of the antagonists [^{3}H] SCH23390 and [^{3}H] Spiperone to D1- and D2-DA receptors, respectively, in the STR and NAcc of rats treated with Glyph, measured 2 days after treatment ended. Specific binding to the D1-DA receptor in STR (A) and NAcc (C), and to the D2-DA receptor in STR (B) and NAcc (D). Absolute values (nmol/mg protein ± SE) for vehicle groups: (A) 592.36 ± 48.93, (B) 225.78 ± 13.96, (C) 817.13 ± 32.70, (D) 150.54 ± 9.02; n = 6. *Different from vehicle, $p < 0.05$.

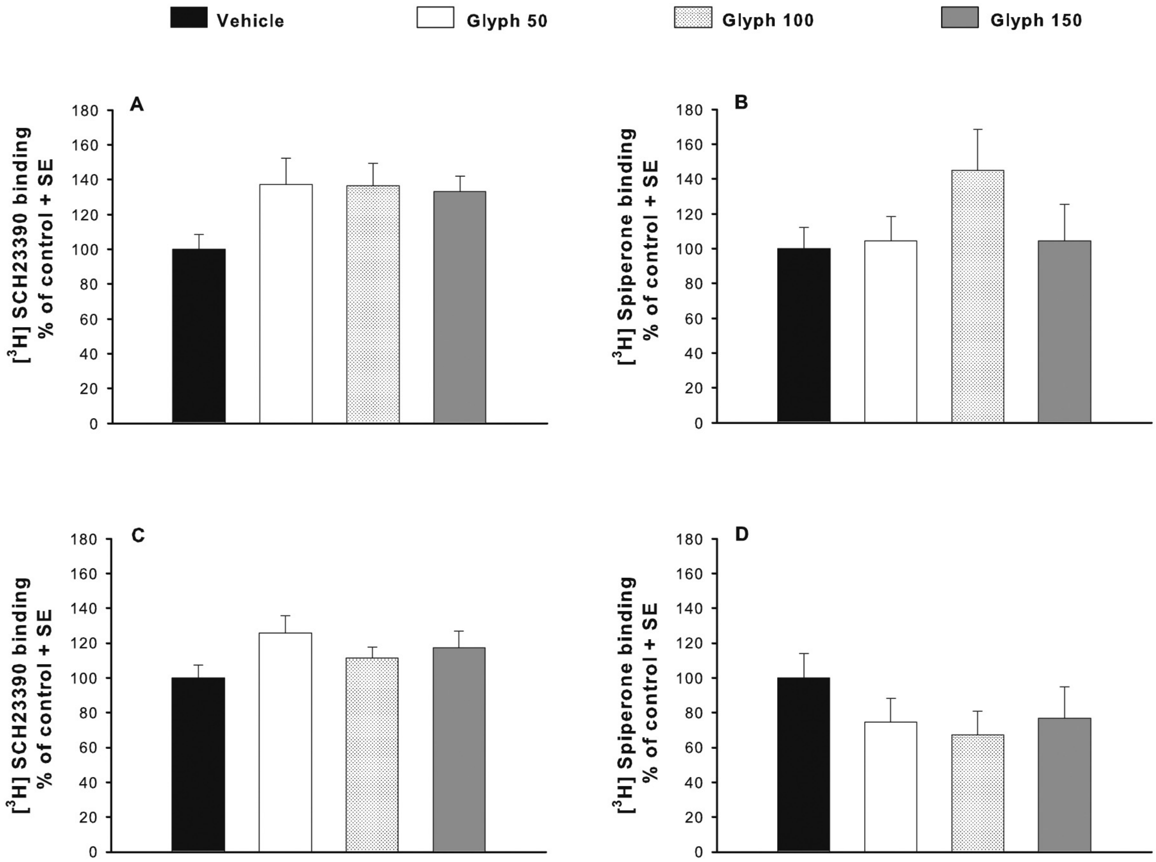

Fig. 6. Specific binding of the antagonists [^{3}H] SCH23390 and [^{3}H] Spiperone to D1- and D2-DA receptors, respectively, in STR and NAcc of rats treated with Glyph, measured 16 days after treatment ended. Specific binding to D1-DA receptors in STR (A) and NAcc (C), and to D2-DA receptors in STR (B) and NAcc (D). Absolute values (nmol/mg protein $\pm$ SE) for vehicle groups: (A) 259.38 $\pm$ 21.66, (B) 138.03 $\pm$ 16.61, (C) 333.06 $\pm$ 24.20, (D) 62.71 $\pm$ 8.64. n = 6.

3.1.7. Repeated Glyph exposure does not alter the number of TH-positive cells in the substantia nigra or the ventral tegmental area 2 and 16 days post-treatment

When measured 2 and 16 days after treatment, the number of TH-positive cells in the three subregions of the substantia nigra (SNpc, SNpl and SNpr) or in the four regions of the VTA (VTA, PBP, PN and RLi) was similar in all groups (Table 4).

3.2. Experiment 2. Effects of acute exposure to Glyph (150 mg/kg BW) on DA release in the STR and NAcc

As shown in Fig. 8A, acute 150 mg Glyph/kg BW administration (i.p.) not only decreased striatal extracellular DA levels but also reduced striatal DA release induced by the perfusion of a high-potassium (60 mM) solution. Striatal DA release levels showed group, sample and interaction effects (treatment $\times$ time) time) ($F_{(1,9)} = 5.35$, $p = 0.04$; $F_{(14,126)} = 40.32$, $p < 0.0001$; $F_{(14,126)} = 4.96$, $p < 0.0001$, respectively), while striatal DOPAC and HVA levels only showed significant sample effects ($F_{(14,126)} = 10.42$–15.82, $p < 0.0001$). In particular, striatal extracellular DA levels were decreased when measured 20 min, and 2 h after Glyph injection (sample 4: $t_{(9)} = 2.76$, $p = 0.02$; sample 9: $t_{(9)} = 2.561$, $p = 0.03$), but DA levels in samples 5 and 6 decreased non-significantly ($t_{(9)} = 2.05$, 1.93, $p < 0.1$, respectively) (Fig. 8). Striatal DA release induced by the infusion of a high-potassium solution decreased in samples 13 and 15 in the 150 mg Glyph group in comparison to the vehicle group ($t_{(9)} = 2.35$, 2.40, $p < 0.05$, respectively). After the high-potassium challenge, in sample 18, striatal extracellular DA levels were lower in the Glyph 150 group ($t_{(9)} = 3.13$, $p = 0.01$).

In contrast to the results in STR, levels of DA, DOPAC and HVA in NAcc dialysates were similar in both groups, even during the challenge with high-potassium (60 mM) ringer (Fig. 9). Only significant sample effects for DA, DOPAC and HVA levels ($F_{(14,140)} = 15.89$–30.36, $p < 0.0001$) were observed in NAcc.

4. Discussion

In the present study, we have shown that repeated Glyph exposure causes hypoactivity, which is not the result of discomfort or general malaise, since Glyph-exposed rats did not show taste aversion or changes in place preference (Supplementary Figs. 2–4). In addition, although each rat received a total of approximately 270 mg of Glyph at the end of treatment for the higher doses (45 mg Glyph in each injection for the doses of 150 mg/kg), there were no alterations in body weight or in enzymatic markers of hepatic, kidney or heart damage, and there was no mortality as a result of Glyph treatment.

Although the intended targets of Glyph are plants, exposure to this herbicide affects motor behavior and the dopaminergic system of rats and humans. Indeed, after exposure to commercial mixtures that contain Glyph, Parkinsonian-like symptoms in human subjects have been observed (Barbosa et al., 2001; Wang et al., 2011). In the present study the most severe effects of Glyph on locomotor activity were observed immediately after each administration. Hypoactivity also was observed in total distance and vertical activity 48 h after each Glyph injection and even when the animals received saline 2 days after treatment ended.

Similar short-term effects, for 5 days after the last administration, have been observed in male rats repeatedly exposed to ATR

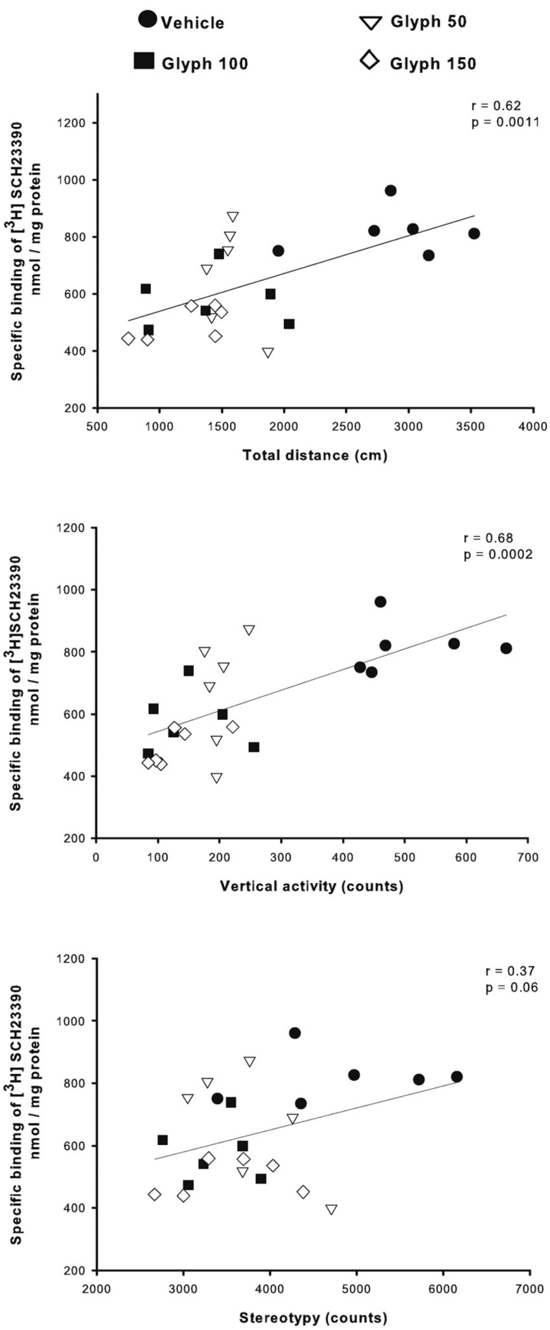

Fig. 7. Correlation between the specific binding to D1-DA receptor in NAcc measured 2 days after treatment ended, and the mean locomotor activity for the 3-h period following the seven injections (Glyph and vehicle). Correlation coefficients and significance level are shown at the upper right of each panel. The correlations refer to the results obtained from additional groups treated specifically for this purpose according to the experimental design shown in the upper panel of Fig. 1. The locomotor activity of these groups was similar to that of the other groups reported in this paper (data not shown).

(Rodríguez et al., 2013). Other herbicides, such as paraquat (Brooks et al., 1999) or 2,4-dichlorophenoxyacetic acid (2,4-D) (Schulze and Dougherty, 1988) have also been shown to affect the motor behavior in rodent models.

Importantly, in the present study we found that D1-DA receptor binding was reduced in the NAcc, and this reduction showed a positive correlation with the decrease observed in ambulatory activity recorded immediately after Glyph administration. Our findings are consistent with studies that have shown that both the NAcc and D1-DA receptors are involved in motor control. In this regard, the intracerebral administration of 1 or 2 μg/side of the D1-antagonist SCH23390 into the core or shell of the rat NAcc decreases crossings in an open field (Baldo et al., 2002), while acute or chronic subcutaneous administration of SCH23390 (0.5 mg/kg) into male rats causes catalepsy (Hess et al., 1988).

Table 4
Number of tyrosine hydroxylase-positive cells in substantia nigra and ventral tegmental area of glyphosate-treated rats measured 2 and 16 days post-treatment.

	Brain area			
	SNpc	SNpl	SNpr	VTA
2 days post-treatment				
Vehicle	102 ± 22	24 ± 6	26 ± 5	149 ± 39
Glyph 50	97 ± 25	19 ± 2	27 ± 5	116 ± 28
Glyph 100	143 ± 28	22 ± 3	33 ± 6	164 ± 31
Glyph 150	137 ± 39	22 ± 3	30 ± 6	147 ± 34
16 days post-treatment				
Vehicle	131 ± 11	21 ± 2	32 ± 2	169 ± 18
Glyph 50	173 ± 29	25 ± 3	41 ± 3	212 ± 33
Glyph 100	125 ± 15	21 ± 3	28 ± 3	165 ± 17
Glyph 150	125 ± 17	21 ± 3	31 ± 3	155 ± 17

Values are mean ± SE. n = 5–7. SNpc, substantia nigra pars compacta; SNpl, substantia nigra pars lateralis; SNpr, substantia nigra pars reticulata; VTA, ventral tegmental area.

Additionally we observed that striatal extracellular DA levels decreased by 50% in the first microdialysis sample (20 min) obtained after Glyph (150 mg) administration. This decrease coincided with the 50% reduction in total distance traveled observed during the first 15 min after Glyph (150 mg) administration in the behavioral experiments. In addition, striatal DA levels (both basal release and induced by high potassium) were significantly decreased after Glyph (150 mg) administration for up to 5 h. Thus, the decrease in striatal extracellular DA levels after acute Glyph administration could, in principle, explain the behavioral hypoactivity (Supplementary Fig. 1).

The effects of Glyph exposure on locomotor activity and dopamine markers were evident for 48 h after the end of exposure but not after 16 days, suggesting that these alterations occur while a "critical" concentration of Glyph is present in the system. With regard to clearance of Glyph, in male rats orally exposed to 10 mg of [^{14}C] and [^{12}C] Glyph/kg BW, approximately 76% of Glyph is eliminated through urine and feces within 28 h after administration, and only about 1% of the Glyph injected remains in the animal 7 days after exposure (Brewster et al., 1991).

Since the effects on D1-DA receptors in the NAcc were only detected at 2 days but not at 16 days post-treatment, we concluded that repeated Glyph exposure causes short-term effects on mechanisms that regulate either the density or the affinity of D1-DA receptors in NAcc. Direct mechanisms could involve modifications ranging from transcription to translocation of receptors into the membrane or structural changes of the receptor. For example, the herbicide 2,4-D has been shown to cause internalization of D2-DA receptors, altering their density in striatum, prefrontal cortex, hippocampus and cerebellum of rats chronically exposed to 70 mg 2,4-D/kg (Bortolozzi et al., 2002). An indirect mechanism could involve temporal alterations of membrane features that affect receptor properties. Cattani et al. (2014) detected increases in levels of TBARS, a lipoperoxidation marker, in hippocampal slices of immature rats exposed to 0.01% Roundup®, an herbicide which contains Glyph. Similarly, in male rats exposed to Glyph, Astiz et al. (2009) found decreases in cardiolipin levels, a membrane glycerolipid, in the mitochondria of the substantia nigra and cerebral cortex. Also, the mixture of Glyph with other pesticides increased TBARS levels in the mitochondrial fraction of substantia nigra and cerebral cortex. In the present study, we did not determine whether Glyph exposure affected the density or the affinity of DA receptors.

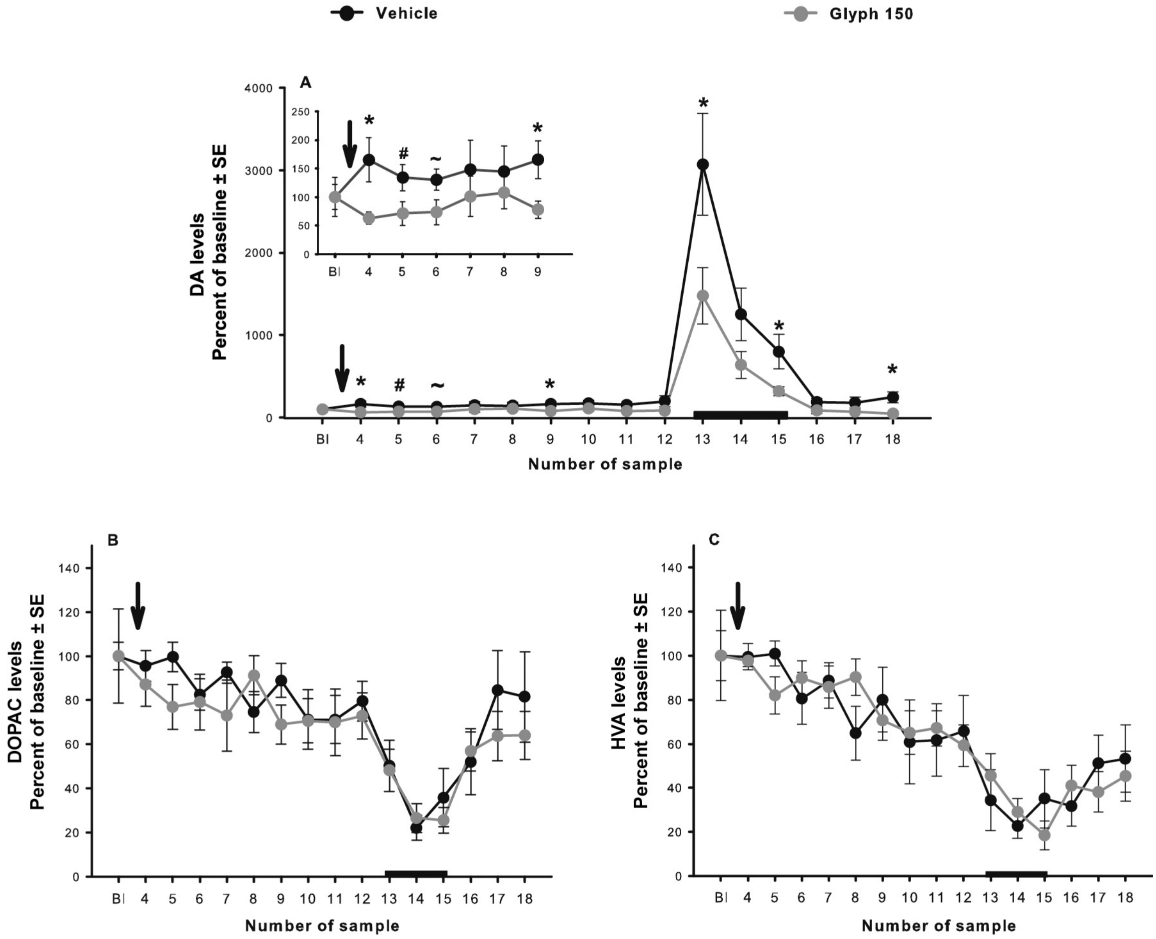

Fig. 8. Acute effects of Glyph on release of DA and its metabolites in STR of anesthetized rats. Extracellular levels of DA (A), DOPAC (B) and HVA (C). Arrow shows the time of injection with vehicle or 150 mg Glyph/kg BW. Dark bar indicates the time of high-potassium infusion (60 nM). n = 5–6. Inset in (A) shows an expanded scale of the results from baseline (Bl) to sample 9. Number of sample corresponds to each of the 18 samples collected every 20 min, the first three were considered as baseline. Absolute values of baseline (ng/mL ± SE) for vehicle group: DA = 0.2 ± 0.04, DOPAC = 67 ± 4, HVA = 71 ± 8, and treated group: DA = 0.3 ± 0.1, DOPAC = 44 ± 9, HVA = 53 ± 10. *Different from vehicle group $p < 0.05$; $^{\#}p$ = 0.05–0.07; $^{\sim}p < 0.1$.

The mechanisms by which acute Glyph administration immediately decreases extracellular levels of basal striatal DA are unknown, but we propose the following: (1) that Glyph may accelerate DA reuptake through an overactive dopamine transporter, (2) that mobilization of DA-containing vesicles to the extracellular membrane, and/or vesiculation are decreased, (3) that Glyph may affect DA synthesis, since this decrease is still present 3-h after Glyph administration, and (4) that Glyph may interfere with the phosphorylation of TH, mainly at Ser 40 or 19, since TH activation by phosphorylation is the primary short-term mechanism for maintenance of catecholamine levels (Dunkley et al., 2004). The same mechanisms may be responsible for the decrease in high-potassium-induced DA release.

Acute Glyph administration did not modify striatal DOPAC or HVA levels before or after the high-potassium challenge. A similar observation was also made in male rats exposed to the pesticides Allethrin (20 or 60 mg/kg i.p.) and Cyhalothrin (20 or 60 mg/kg i.p.) (Mubarak Hossain et al., 2006). Both pesticides decreased DA release; they did not alter DOPAC or HVA levels but did increase 3-methoxytyramide (3-MT) levels. This finding suggests that decreased DA levels may be the result of the metabolism of DA to 3-MT (Mubarak Hossain et al., 2006). Although these are possible explanations for our results, they remain to be tested in the case of Glyph.

Although the hypoactivity caused by acute Glyph administration is similar to that observed after administration of dopamine antagonists, Glyph may affect additional neurotransmitter systems involved in motor control. Recently, it was reported that hippocampal slices of 15-day-old rats chronically exposed to 1% Roundup® showed a decrease in [^{3}H]-glutamate reuptake and an increase in glutamate release (Cattani et al., 2014). In rodent models, it has been shown that herbicides such as 2,4-D (Bortolozzi et al., 2003), ATR (Lin et al., 2013) or paraquat (Kuter et al., 2007) affect several neurotransmitters systems such as noradrenaline, serotonin or their metabolites in cerebellum, prefrontal cortex, striatum, midbrain, substantia nigra or cerebellum. In the present study, we found that serotonin and its metabolite levels were not affected by Glyph administration (Table 3).

Our results show that the STR and NAcc were differentially vulnerable to Glyph exposure. This may be due to the peculiar anatomic and physiological properties of the nigrostriatal and mesolimbic pathways such as brain connectivity (Gerfen, 2004; Humphries and Prescott, 2010), membrane properties, cellular excitability, lipoperoxidation levels, and enzymatic activity of catalase and glutathione peroxidase (Hung and Lee, 1998; Ma et al., 2012), among others. In this study, ambulation and rearing behavior correlated with D1-DA binding in the NAcc but not in the STR; on the other hand, stereotypical behavior, which is considered to be controlled by the STR, was not as affected and did not correlate with DA binding. Thus, this suggests a differential sensitivity to Glyph in these nuclei.

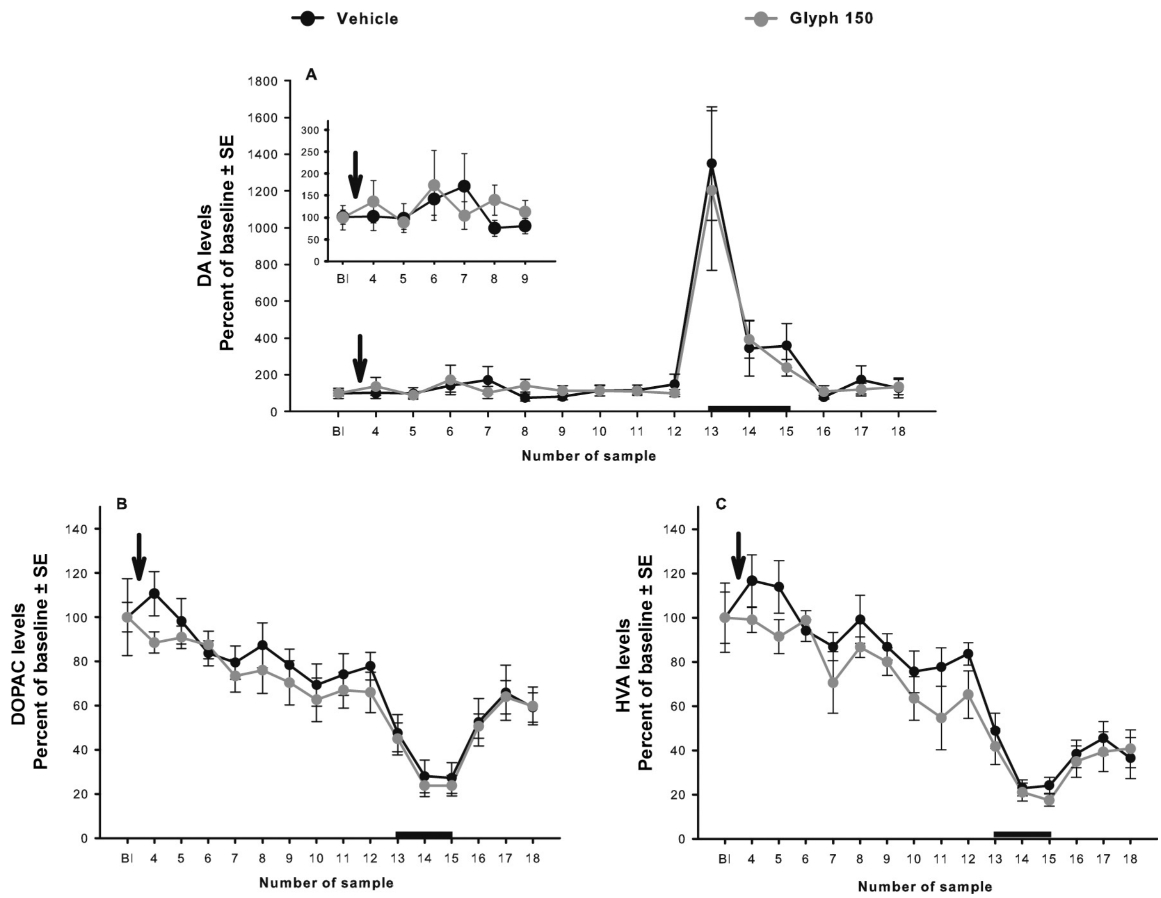

Fig. 9. Acute effects of glyphosate injection on release of DA and its metabolites in NAcc of anesthetized rats. Extracellular levels of DA (A), DOPAC (B) and HVA (C). Arrow shows the injection with vehicle or 150 mg Glyph/kg BW. Dark bar indicates the high-potassium infusion (60 nM). $n = 6$. Inset in (A) shows the results from baseline (Bl) to sample 9 on an expanded scale. Number of sample corresponds to each of the 18 samples collected every 20 min, the first three were considered as baseline. Absolute values of baseline (ng/mL ± SE) for vehicle group: DA = 0.06 ± 0.01, DOPAC = 30 ± 5, HVA = 43 ± 7, and treated group: DA = 0.08 ± 0.02, DOPAC = 32 ± 2, HVA = 46 ± 5.

5. Conclusion

In the present study, we have shown that exposure to Glyph, the active ingredient employed in some herbicides, decreased locomotor activity, binding to the D1-DA receptor in the NAcc, and extracellular DA levels in the STR. These results suggest that Glyph affects the dopaminergic system. Additional experiments are necessary in order to unveil the specific targets of Glyph on dopaminergic system, and whether Glyph could be affecting other neurotransmitter systems involved in motor control.

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Conflict of interest

The authors declare that there are no conflicts of interest.

Transparency document

The Transparency document associated with this article can be found in the online version.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.neuro.2014.12.001.

References

Astiz M, de Alaniz JTM, Marra CA. Effect of pesticides on cell survival in liver and brain rat tissues. Ecotoxicol Environ Saf 2009;72:2025–32.

Barbosa E, Da costa M, Bacheschi L, Scaff M, Leite CC. Parkinsonism after glycine-derivate exposure. Mov Disord 2001;16:565–8.

Baldo BA, Sadeghian K, Basso A, Kelley AE. Effects of selective dopamine D1 or D2 receptor blockade within nucleus accumbens subregions on ingestive behavior and associated motor activity. Behav Brain Res 2002;137:165–77.

Bardullas U, Giordano M, Rodríguez VM. Atrazine is primarily responsible for the toxicity of long-term exposure to a combination of atrazine and inorganic arsenic in the nigrostriatal system of the albino rat. Neurotoxicol Teratol 2013;40:59–66.

Bortolozzi A, Duffard R, Antonelli M, Evangelista de Duffard AM. Increased sensitivity in dopamine D2-like brain receptors from 2,4-dichlorophenoxyacetic acid (2,4-D)-exposed and amphetamine-challenged rats. Ann N Y Acad Sci 2002;965:314–23.

Bortolozzi A, Duffard R, de Duffard AM. Asymmetrical development of the monoamine systems in 2,4-dichlorophenoxyacetic acid treated rats. Neurotoxicology 2003;24:149–57.

Brewster D, Warren J, Hopkins W. Metabolism of glyphosate in Sprague-Dawley rats: tissue distribution, identification, and quantification of glyphosate-derived materials following a single oral dose. Fundam Appl Toxicol 1991;17:43–51.

Brooks AI, Chadwick C, Gelbard A, Cory-Slechta D, Federoff HJ. Paraquat elicited neurobehavioral syndrome caused by dopaminergic neuron loss. Brain Res 1999;823:1–10.

Cattani D, de Liz Oliveira Cavalli VL, Heinz Rieg CE, Domingues JT, Dal-Cim T, Tasca CI, et al. Mechanisms underlying the neurotoxicity induced by glyphosate-based herbicide in immature rat hippocampus: involvement of glutamate excitotoxicity. Toxicology 2014;320:34–45.

Daruich J, Zirulnik F, Gimenez MS. Effect of the herbicide glyphosate on enzymatic activity in pregnant rats and their fetuses. Environ Res 2001;85:226–31.

Dill GM, Cajacob CA, Padgette SR. Glyphosate-resistant crops: adoption, use and future considerations. Pest Manag Sci 2008;64:326–31.

Dunkley PR, Bobrovskaya L, Graham ME, Von Nagy-Felsobuki EI, Dickson PW. Tyrosine hydroxylase phosphorylation: regulation and consequences. J Neurochem 2004;91:1025–43.

Gerfen C. Basal ganglia. In: Paxinos G, editor. In the rat nervous system. San Diego, CA, USA: Elsevier Academic Press; 2004, pp. 455–508.

Gui YX, Fan XN, Wang HM, Wang G, Chen SD. Glyphosate induced cell death through apoptotic and autophagic mechanisms. Neurotoxicol Teratol 2012;34:344–9.

Hess EJ, Norman AB, Creese I. Chronic treatment with dopamine receptors antagonist: behavioral and pharmacologic effects on D1 and D2 dopamine receptors. J Neurosci 1988;8:2361–70.

Hoffman G, Le WW, Sita LV. The importance of titrating antibodies for immunocytochemical methods. In: Gerfen CR, Holmes A, Sibley D, Skolnick P, Wray S, editors. Current protocols in neuroscience. John Wiley & Sons, Inc.; 2008 , pp. 2.12.1–2.12.26.

Mubarak Hossain M, Suzuki T, Sato N, Sato I, Takewaki T, Suzuki K, et al. Differential effects of pyrethroid insecticides on extracellular dopamine in the striatum of freely moving rats. Toxicol Appl Pharmacol 2006;217:25–34.

Humphries MD, Prescott T. The ventral basal ganglia, a selection mechanism at the crossroads of space, strategy and reward. Prog Neurobiol 2010;90:385–417.

Hung HC, Lee EH. MPTP produces differential oxidative stress and antioxidative responses in the nigrostriatal and mesolimbic dopaminergic pathways. Free Radic Biol Med 1998;24:76–84.

Kuter K, Smialowska M, Wieronska J, Zieba B, Wardas J, Pietraszek M, et al. Toxic influence of subchronic paraquat administration on dopaminergic neurons in rats. Brain Res 2007;1155:196–207.

Levant B. Characterization of dopamine receptors. In: Enna SJ, Williams M, Kenakin T, McGonigle P, Ruggeri B, editors. Current protocols in pharmacology. John Wiley & Sons, Inc.; 2007, pp. 1.6.1–1.6.15.

Lin Z, Dodd CA, Filipov NM. Short-term atrazine exposure causes behavioral deficits and disrupt monoaminergic systems in male C57BL/6 mice. Neurotoxicol Teratol 2013;39:26–35.

Ma YY, Cepeda C, Chatta P, Franklin L, Evans CJ, Levine MS. Regional and cell-type-specific effects of DAMGO on striatal D1 and D2 dopamine receptor-expressing medium-sized spiny neurons. ASN Neuro 2012;4:59–70.

Malhotra RC, Ghia DK, Cordato DJ, Beran RG. Glyphosate-surfactant herbicide-induced reversible encephalopathy. J Clin Neurosci 2010;17:1472–3.

Menkes DB, Temple WA, Edwards IR. Intentional self-poisoning with glyphosate containing herbicides. Hum Exp Toxicol 1991;10:103–7.

Pan Y, Berman Y, Haberny S, Meller E, Carr KD. Synthesis, protein levels, activity, and phosphorylation state of tyrosine hydroxylase in mesoaccumbens and nigrostriatal dopamine pathways of chronically food-restricted rats. Brain Res 2006;1122:135–42.

Paxinos G, Watson C. The brain rat in stereotaxic coordinates. 5th ed. San Diego, CA, USA: Elsevier Academic Press; 2005.

Rodríguez VM, Limon-Pacheco J, Mendoza-Trejo MS, Gonzalez-Gallardo A, Hernández-Plata I, Giordano M. Repeated exposure to the herbicide atrazine alters locomotor activity and the nigrostriatal dopaminergic system of the albino rat. Neurotoxicology 2013;34:82–94.

Sato C, Kamijo Y, Yoshimura K, Ide T. Aseptic meningitis in association with glyphosate-surfactant herbicide poisoning. Clin Toxicol 2011;49:118–20.

Schulze GE, Dougherty J. Neurobehavioral toxicity and tolerance to the herbicide 2,2-dichlorophenoxyacetic acid-*n*-butyl ester (2,4-D ester). Fundam Appl Toxicol 1988;10:413–24.

Thiruchelvam M, Richfield EK, Baggs RB, Tank AW, Cory-Slechta DA. The nigrostriatal dopaminergic system as a preferential target of repeated exposures to combined paraquat and maneb: implications for Parkinson's disease. J Neurosci 2000;20: 9207–14.

Wang G, Fan XN, Tan YY, Cheng Q, Chen SD. Parkinsonism after chronic occupational exposure to glyphosate. Parkinsonism Relat Disord 2011;17:486–7.

Williams G, Kroes R, Munro IC. Safety evaluation and risk assessment of the herbicide roundup and its active ingredient, glyphosate, for humans. Regul Toxicol Pharmacol 2000;31:117–65.